

Section D - PLC I/O

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Description

PLC functions.

Topics

Name	Description
c:wd_find_sel_plc	Returns list of PLC database "data" to be used by the (c:wd_inplc_n routine to insert a PLC module.
c:wd_inplc	Invoke PLC selection dialog box. Pick and insert PLC module per use prompts.
c:wd_inplc_altdat	Invoke PLC selection dialog box, supply database. Use "datfnam" in of defaulting to "ace_plc.mdb" for manufacturer file selection.
c:wd_inplc_nd	Insert PLC module. Starts at "plc_xy". Allows partial module to be inserted (i.e. splitting long module into multiple inserts).
c:wd_inplc_nd2	Simplified PLC insert from all command line params, whole module inserts.
c:wdio_autorun	Start Spreadsheet-to-PLC I/O generation utility. Invoke AutoCAD Eletrical's Spreadsheet-to-PLC I/O generation utility from the comm line. The "wdio.lsp" file must be in memory ("apload") for this to r

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Description

Returns list of PLC database "data" to be used by the (c:wd_inplc_nd...) routine to insert a PLC module.

AutoLISP

(c:wd_find_sel_plc catkey)

Parameters

Example

```
(setq data (c:wd_find_sel_plc "1771-IA"))
```

Module

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_inplc

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Description

Invoke PLC selection dialog box. Pick and insert PLC module per user prompts.

AutoLISP

```
(c:wd_inplc)
```

Module

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_inplc_altdat

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Description

Invoke PLC selection dialog box, supply database. Use "datfnam" instead of defaulting to "ace_plc.mdb" for manufacture selection.

Parameters

Parameters	Description
datfnam	List of vendor names/parametric database.

Module

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c:wd_inplc_nd

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Description

Insert PLC module. Starts at "plc_xy". Allows partial module to be inserted (i.e. splitting long module into multiple insert

AutoLISP

(c:wd_inplc_nd plc_xy style firstaddr veryfirsta base_carry beg_ix end_ix usr_rv orientation data a

Parameters

Parameters	Description
plc_xy	Insertion point.
style	PLC "style" number ("1" through "9"). Defines the appearance of the graphics that will be generated.
firstaddr	First addr for module or this piece of broken module (o of starting addresses for "combo" module).
veryfirsta	First addr of the overall module when dealing with inse of a split piece. Nil= no split or use "firstaddr" value.
base_carry	Numbering method (8 = octal, 10 = decimal, 16 = hex prompt user if necessary).
beg_ix	1
end_ix	Count of last addr point (for example, 16 or 32 or any integer number).

data	List of data returned by wd_find_sel_plc.
alst	nil if no "prompt responses", or (list "" "%%1 prompt answer" "%%2 prompt answer" ...)

Example 1

Insert AB 1771-IAD.

```
(c:wd_inplc_nd (getpoint "PLC ins pt ") "1" (getstring "addr=") nil nil 1 99 nil "V"
(c:wd_find_sel_plc "1771-IAD")
(list "" (getstring " rack=") "" (getstring "slot=")))
```

Example 2

Insert AB 1771-IAD with "unused/extra" option enabled.

```
(setq data (c:wd_find_sel_plc "1771-IAD"))
(setq data (list (nth 0 data) (nth 2 data) (nth 1 data) (nth 3 data) (nth 4 data)))
(c:wd_inplc_nd (getpoint "PLC ins pt ") "1" (getstring "addr=") nil nil 1 99 nil "V" data (list ""
(getstring "
rack=") "" (getstring "slot=")))
```

Example 3

Insert bottom 8-point half of a split AB 1771-IAD module.

```
(c:wd_inplc_nd (getpoint "PLC ins pt ") "1" (getstring "addr=") nil nil 9 99 nil "V"
(c:wd_find_sel_plc "1771-IAD")
(list "" (getstring " rack=") "" (getstring "slot=")))
```

Example 4

Combo module with two starting addressed passed (first for outputs, 2nd for inputs).

```
(c:wd_inplc_nd (getpoint "PLC ins pt ") "1" (list "O:0010" "I:10010") nil nil 1 99 nil "V"
(c:wd_find_sel_plc "1746
-I012") (list "" (getstring " rack=") "" (getstring "slot=")))
```

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AutoCAD Electrical 2025: AutoLISP Reference

c:wd_inplc_nd2

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AutoLISP

```
(c:wd_inplc_nd2 param_lst)
```

Parameters

Parameters	Description
param_lst	(nth 0) = part number (the "K=" match in the plc*.dat (nth 1) = plc_xy insertion point. (nth 2) = style number "1" = style 1. (nth 3) = starting address (or list of starting addresses "combo" module). (nth 4) = base_carry (10 = decimal, 16 = hex, 8 = oct (nth 5) = orientation "H" = horizontal module, "V" = v (default).

Returns

4 element list where (nth 3) = ent name of new PLC module inserted. Nil on failure.

Example 1

```
(c:wd_inplc_nd2 (list "1771-IAD" (list 10.0 12.0 0.0) "2" "I:10010" 10 "V"))
```

Example 2

Combo module with two starting addresses passed (first for outputs, 2nd for inputs.

```
(c:wd_inplc_nd2 (list "1746-IO12" (list 10.0 12.0 0.0) "2" (list "O:0010" "I:10010") 10 "V"))
```

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Description

```
(c:wdio_autorun param_lst)
```

Parameters

Parameters	Description
param_lst	(nth 0) = xlsfnam = spreadsheet file name (.xls, .csv, (nth 1) = wdi_fnam = settings file name (nil = use def (nth 2) = continuation_dwglst (nil = increment starting file name for each new drawing required to handle the spreadsheet's data. (nth 3) = optional XLS or MDB sheet/table name (nil = "Sheet1" or first table found with data). (nth 4) = Starting drawing file name (nil=use current : drawing " must be a "named" drawing) (nth 5) = "other_options": nil=none; (1's bit set = pau display main dialog box)

Example

```
(if (not (boundp 'c:wdio_autorun))
  (progn ;wdio program not yet in memory. Try to find source file and load it.
    (setq x (findfile "wdio.lsp"))
    (if (AND (not x) GBL_wd_sup)
      (setq x (findfile (strcat GBL_wd_sup "wdio.lsp"))))
    (if (not x)
      (alert "Cannot find wdio.lsp to loadnApload it before restarting this program.")
      ;ELSE
      (load x); load the wdio.lsp program into memory
    )
  )
)
(if (boundp 'c:wdio_autorun)
  (progn
    (c:wdio_autorun (list (c:ace_find_file "demopl.c.xls" 16) nil nil nil "c:/files/PLC01.dwg" 1))
  )
)
```

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